

Replication Report: *Magnetic control of spin-orbit fields: a first-principles study of Fe/GaAs junctions*

Replication (subagent texture-gmitra2013)
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We replicate the symmetry-based spin-orbit-field (SOF) framework of Gmitra, Matos-Abiague, Draxl and Fabian [PRL **111**, 036603 (2013); arXiv:1303.2510]. The paper’s DFT (relativistic FLAPW/FLEUR on Fe/GaAs slabs) is out of scope for in-process replication; instead we reimplement the paper’s analytic core—the C_{2v} SOC Hamiltonian, the Bychkov–Rashba/Dresselhaus decomposition, the symmetry extraction formulas (Eqs. 4–9), and the magnetization-angle parametrization of Table I—and verify the internal consistency and the downstream physical claims. All six machine-checkable claims pass. In particular we reproduce (i) the exact self-consistency of the extraction method, (ii) the sign change of $\alpha_1(\theta)$, (iii) the sign flip of $\alpha_1\beta_1$ between $[1\bar{1}0]$ and $[110]$ (band $n=2$ does not flip), (iv) the linear-in- k regime and anisotropic “butterfly” textures, and (v) the pure $\cos(2\theta)$ C_{2v} angular form with band 1 far more magnetization-sensitive than band 2.

CENTRAL CLAIMS

The paper’s headline results are:

1. SOFs at the C_{2v} Fe/GaAs(001) interface depend not only on the interface electric field but *surprisingly strongly on the Fe magnetization orientation*—enabling magnetic control.
2. The k-space SOF patterns are highly anisotropic and change qualitatively with band/energy (“butterflies”); the anisotropy axes can be *flipped* by rotating the magnetization.
3. A symmetry-based method extracts $\mathbf{w}(\mathbf{k})$ (magnitude and direction) for a generic Bloch state directly from ab-initio bands, valid also at large k (not just near Γ).

MODEL

The most general C_{2v} -allowed in-plane SOC Hamiltonian (Eq. 1) is

$$H_{\text{so}} = \mu_n(k_x, k_y, \theta) k_x \sigma_y + \eta_n(k_x, k_y, \theta) k_y \sigma_x, \quad (1)$$

with $x \parallel [1\bar{1}0]$, $y \parallel [110]$, θ the magnetization angle from $[1\bar{1}0]$, and (Eq. 2) $\mu_n = \mu_n^{(0)} + \mu_n^{(1)} k_x^2 + \dots$, $\eta_n = \eta_n^{(0)} + \eta_n^{(1)} k_x^2 + \dots$. Writing $H_{\text{so}} = \mathbf{w}_n(\mathbf{k}) \cdot \boldsymbol{\sigma}$ (Eq. 3), $\mathbf{w}_n = (\eta_n k_y, \mu_n k_x, 0)$. Near Γ , $\alpha_n = (\mu_n^{(0)} - \eta_n^{(0)})/2$ (Bychkov–Rashba) and $\beta_n = (\mu_n^{(0)} + \eta_n^{(0)})/2$ (Dresselhaus). The angular dependence (Table I, Eqs. 10–11) is

$$\alpha_n(\theta) \simeq A_n^{(+)} + B_n^{(+)} \cos(2\theta), \quad (2)$$

$$\beta_n(\theta) \simeq A_n^{(-)} + B_n^{(-)} \cos(2\theta), \quad (3)$$

with the paper’s Table I coefficients (meV Å) used verbatim.

The symmetry extraction (Eqs. 4–7) reads $w_{nx} = \sigma [\Delta E_n^{\text{so}} + \Gamma_n^{\text{so}}]/(2 \cos \theta)$, $w_{ny} = \sigma [\Delta E_n^{\text{so}} - \Gamma_n^{\text{so}}]/(2 \sin \theta)$,

with $\Delta E_n^{\text{so}} = [E_n(\mathbf{k}) - E_n(-\mathbf{k})]/2$ and $\Gamma_n^{\text{so}} = [E_n(-k_x, k_y) - E_n(k_x, -k_y)]/2$; the near- Γ parameters follow from band velocities a_n, b_n via Eqs. (8–9).

METHOD (THIS WORK)

We implement the above in `code/sof_model.py` and verify claims in `code/run_analysis.py`. We build a toy band $E_n(\mathbf{k}, \theta) = c(k_x^2 + k_y^2) + \mathbf{w}_n(\mathbf{k}) \cdot \hat{\mathbf{m}}$ (first-order in-plane perturbation theory, $\hat{\mathbf{m}} = (\cos \theta, \sin \theta)$) and feed it back through Eqs. (4–9) to confirm the extraction is exact. We take Table I as the ab-initio ground truth and check the physical consequences the paper draws from it.

RESULTS

All results in `results/metrics.json`; figures in `figs/`.

- **C1 (extraction round-trip).** Eqs. (4–9) recover the input (α, β) to $< 10^{-6}$ meV Å and $w_{x,y}$ to $< 10^{-9}$ meV across bands $n=1-4$ and nine test angles. The method is exact by construction.
- **C2 (α_1 sign change).** $\alpha_1(\theta) \in [-6.68, +5.84]$ meV Å crosses zero: magnetic control confirmed.
- **C3 (product sign flip).** $\alpha_1\beta_1$: +46.8 at $[1\bar{1}0]$, −91.3 at $[110]$ (flips). $\alpha_2\beta_2$: +2419 vs +2502 (no flip). Matches the paper exactly—band 1 axes flip, band 2 preserved.
- **C4 (linearity + anisotropy).** $|w|/k$ is k -independent in the linear regime (rel. dev. $< 10^{-9}$). On the $k = \pi/8d$ contour the max/min of $|w|/k$ is 42.6 ($M \parallel [1\bar{1}0]$) vs 2.19 ($M \parallel [110]$): strongly anisotropic butterflies whose shape changes with M .

- **C5 (C_{2v} angular form).** Angular modulation $|B/A|$ for α : band 1 = 14.9 vs band 2 = 0.043—band 1 is $\sim 350\times$ more magnetization-sensitive, matching the paper’s statement.

FIGURES

`fig3_soc_vs_theta.png` reproduces the $E=0$ structure of Fig. 3 (α_n, β_n vs θ). `fig2_sof_butterflies.png` and `fig2_polar_wk.png` reproduce the Fig. 2 SOF vector fields and polar $|w|/k$ on the $\pi/100d, \pi/8d, \pi/5d$ contours for band 1 at $M \parallel [\bar{1}10]$ and $[110]$.

VERDICT

Partial-to-strong replication of the analytic core. The DFT-derived numbers themselves are not reproduced (out of scope), but every symmetry claim and every downstream physical consequence the paper draws from its extracted Table I is reproduced exactly. **Coverage: 6/10** (analytic core + all qualitative claims covered; ab-initio band structure, electric-field sweep, and out-of-plane $\langle s \rangle / \Delta_{xc}$ maps not reproduced). **Agreement: 9/10** (internal consistency and all sign/anisotropy/ordering claims match; capped below 10 because the DFT inputs are taken as given rather than independently computed).